

PROJECT PROPOSAL

Visual Programming

Personal Finance Tracker

A Web Application using ASP.NET MVC with C#

Submitted To:	Miss Shamim
Subject:	Visual Programming
Technology:	ASP.NET MVC C# SQL Server
Project Type:	Semester Project
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1. Introduction

Managing personal finances is one of the most critical life skills, yet many individuals struggle to track their income, expenses, and savings goals effectively. This project proposes the development of a Personal Finance Tracker – a fully-featured web application built using the ASP.NET MVC framework with C# as the primary language.

The application will provide users with a clean, intuitive interface to record financial transactions, categorize spending, set budgets, and visualize trends through charts and reports. By leveraging the Model-View-Controller architectural pattern, the project will demonstrate core concepts of Visual Programming including UI design, data binding, controller logic, and database integration.

2. Project Objectives

The primary objectives of this project are:

- To design and develop a fully functional web-based personal finance tracker using ASP.NET MVC (C#).
 - To implement user authentication and authorization using ASP.NET Identity.
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- To allow users to add, edit, delete, and categorize income and expense transactions.
- To provide a dashboard with real-time financial summaries, charts, and analytics.
- To implement budget management features with alerts when limits are exceeded.
- To generate monthly/yearly financial reports exportable as PDF.
- To demonstrate practical use of Entity Framework Core for database operations.

3. Project Scope

3.1 In Scope

- User registration, login, and profile management.
- Transaction management: add, update, delete income/expense records.
- Category management for transactions (e.g., Food, Rent, Salary, Entertainment).
- Dashboard with summary cards (Total Income, Total Expenses, Net Balance).
- Interactive charts (bar chart, pie chart) for spending breakdown.
- Monthly budget setting with overspend alerts.
- Transaction filtering and search by date, category, or amount.
- Responsive UI using Bootstrap 5 for desktop and mobile compatibility.

3.2 Out of Scope

- Integration with live banking APIs or third-party financial institutions.
- Mobile native applications (iOS/Android).
- Investment portfolio tracking or stock market integration.

4. Key System Features

#	Feature	Description
1	User Authentication	Secure login and registration using ASP.NET Identity with hashed passwords.
2	Transaction Manager	Full CRUD operations for income and expense entries with date and notes.
3	Categories	Custom user-defined categories with icons for classifying transactions.
4	Dashboard	Visual summary of net balance, monthly totals, and spending breakdown charts.
5	Budget Planner	Set monthly spending limits per category with visual progress bars and alerts.
6	Reports & Export	Generate and download monthly financial summaries in PDF format.

7	Search & Filter	Filter transactions by date range, category, type (income/expense), or keyword.
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5. Technical Stack

Layer / Component	Technology Used
Backend Framework	ASP.NET MVC 5 / .NET 6
Programming Language	C#
Frontend	Razor Views, HTML5, CSS3, Bootstrap 5, JavaScript
ORM	Entity Framework Core (Code-First approach)
Database	Microsoft SQL Server / SQL Server Express
Authentication	ASP.NET Core Identity
Charts & Visualization	Chart.js (integrated via Razor View scripts)
IDE	Microsoft Visual Studio 2022
Version Control	Git / GitHub

6. MVC Architecture Overview

The application strictly follows the Model-View-Controller (MVC) design pattern:

- **Model:** Represents the data structure — classes such as Transaction, Category, Budget, and ApplicationUser mapped to SQL Server database tables via Entity Framework Core.
- **View:** Razor (.cshtml) pages that render the UI dynamically based on data passed from controllers. Views use Bootstrap 5 for responsive layout and Chart.js for data visualization.
- **Controller:** Handles HTTP requests, applies business logic, and passes data between Models and Views. Controllers include AccountController, DashboardController, TransactionController, BudgetController, and ReportController.

7. Project Timeline

Week	Phase	Tasks	Deliverable
1	Requirement Analysis & Design	Define requirements, design database schema, wireframe UI screens	SRS Document, ER Diagram, UI Mockups

2	Project Setup & Authentication	Configure ASP.NET MVC project, set up EF Core, implement user login/register	Working authentication module
3	Core Features Development	Build transaction CRUD, categories, and database migrations	Transaction manager module
4	Dashboard & Analytics	Develop dashboard view, integrate Chart.js for charts and summaries	Interactive dashboard
5	Budget & Reports	Budget planner module, PDF report generation, search/filter features	Budget module, export feature
6	Testing & Final Submission	Unit testing, bug fixing, UI polish, documentation, and demo preparation	Final project + presentation

8. Database Design (Key Entities)

The application will use a relational database with the following primary entities:

- **ApplicationUser** — Stores user profile data (UserId, FullName, Email, PasswordHash).
- **Transaction** — Stores financial records (TransactionId, UserId, CategoryId, Amount, Type [Income/Expense], Date, Description).
- **Category** — Defines transaction categories (CategoryId, UserId, Name, Icon, Type).
- **Budget** — Monthly spending limits (BudgetId, UserId, CategoryId, MonthYear, LimitAmount).

Relationships will be implemented using EF Core navigation properties with Code-First migrations for schema management.

9. Expected Outcomes

Upon successful completion, the project will deliver:

1. A fully functioning web application demonstrating ASP.NET MVC architecture in C#.
2. A relational database-backed system with proper CRUD operations.
3. An interactive, responsive user interface with data visualization charts.
4. Secure multi-user support with individual financial data privacy.
5. Complete project documentation including database schema and user manual.
6. A live demo presentation of all major features.

10. Conclusion

The Personal Finance Tracker project is a well-scoped, practical application that directly applies core Visual Programming concepts including MVC architecture, UI design, data binding, and database integration using ASP.NET MVC with C#. It offers real-world value as a tool users can genuinely benefit from, while comprehensively demonstrating the technical skills required by the subject curriculum.

We are confident this project will showcase a strong understanding of web application development principles, and we look forward to your approval and guidance.

Respectfully Submitted

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